Machine Learning-Lecture 19

P1.py

```
C:\ml\L19>python p1.py
   bhk  size_sqft furnishing_status  bathroom    rent
0    1        400         Furnished         1   25000
1    2        600    Semi-Furnished         1   35000
2    3        900         Furnished         2   65000
3    1        350       Unfurnished         1   20000
4    2        750    Semi-Furnished         2   45000
..  ...       ...               ...       ...     ...
88   2        950    Semi-Furnished         2   67000
89   3        990         Furnished         3   99000
90   1        670       Unfurnished         1   34000
91   2        960         Furnished         2   76000
92   3       1000         Furnished         3  100000

[93 rows x 5 columns]
bhk                  0
size_sqft            0
furnishing_status    0
bathroom             0
rent                 0
dtype: int64
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 93 entries, 0 to 92
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   bhk                93 non-null     int64
 1   size_sqft          93 non-null     int64
 2   furnishing_status  93 non-null     object
```

```
 3   bathroom          93 non-null     int64
 4   rent              93 non-null     int64
dtypes: int64(4), object(1)
memory usage: 3.8+ KB
None
    bhk  size_sqft furnishing_status  bathroom
0     1        400         Furnished         1
1     2        600    Semi-Furnished         1
2     3        900         Furnished         2
3     1        350       Unfurnished         1
4     2        750    Semi-Furnished         2
..  ...        ...               ...       ...
88    2        950    Semi-Furnished         2
89    3        990         Furnished         3
90    1        670       Unfurnished         1
91    2        960         Furnished         2
92    3       1000         Furnished         3

[93 rows x 4 columns]
0      25000
1      35000
2      65000
3      20000
4      45000
       ...  
88     67000
89     99000
90     34000
91     76000
92    100000
Name: rent, Length: 93, dtype: int64
```

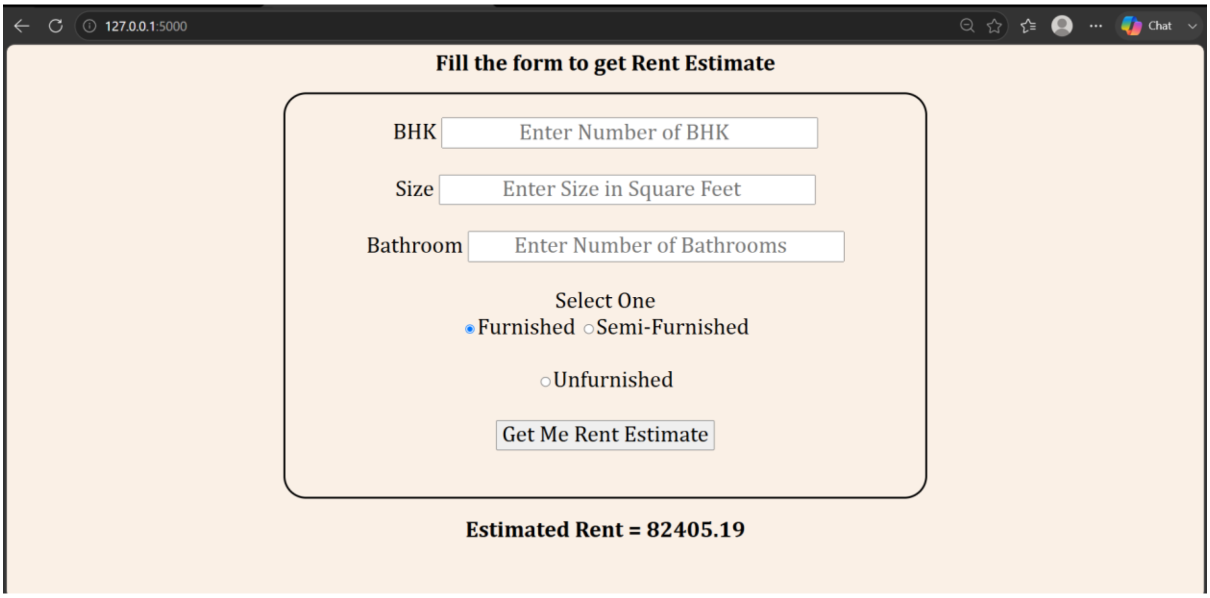
```
   bhk  ...  furnishing_status_Unfurnished
0     1  ...                          False
1     2  ...                          False
2     3  ...                          False
3     1  ...                           True
4     2  ...                          False
..  ...  ...                            ...
88    2  ...                          False
89    3  ...                          False
90    1  ...                           True
91    2  ...                          False
92    3  ...                          False

[93 rows x 6 columns]
0.9595289035414798
model saved
```

P2.py

```
C:\ml\L19>python p2.py
model ready
enter bhk 1
enter size in square feet 10000
enter number of bathrooms 2
1. furnished,2. semi-furnished and 3. unfurnished 1
Estimated Rent = 615785.35
```

P1_rent_app



P4.py

```
C:\ml\L19>python p4.py
   car_name  age_years  kms_driven   price
0    Maruti          2       15000  120000
1    Maruti          1        8000  180000
2    Maruti          3       25000  135000
3    Maruti          4       40000  110000
4    Maruti          5       60000  105000
..      ...        ...         ...     ...
85  Mahindra         7       82000  285000
86  Mahindra         1       11000  475000
87  Mahindra         5       58000  308000
88  Mahindra         3       38000  365000
89  Mahindra        10       95000  260000

[90 rows x 4 columns]
car_name      0
age_years     0
kms_driven    0
```

```
price        0
dtype: int64
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 90 entries, 0 to 89
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   car_name    90 non-null     object
 1   age_years   90 non-null     int64
 2   kms_driven  90 non-null     int64
 3   price       90 non-null     int64
dtypes: int64(3), object(1)
memory usage: 2.9+ KB
None
   age_years  kms_driven  ...  car_name_Maruti  car_name_Tata
0          2       15000  ...             True          False
1          1        8000  ...             True          False
2          3       25000  ...             True          False
3          4       40000  ...             True          False
4          5       60000  ...             True          False
..       ...         ...  ...              ...            ...
85         7       82000  ...            False          False
86         1       11000  ...            False          False
87         5       58000  ...            False          False
88         3       38000  ...            False          False
89        10       95000  ...            False          False

[90 rows x 5 columns]
[[0.11111111 0.10526316 0.         1.         0.        ]
 [0.         0.03157895 0.         1.         0.        ]
 [0.22222222 0.21052632 0.         1.         0.        ]
```

```
[0.33333333 0.36842105 0.         1.         0.        ]
[0.44444444 0.57894737 0.         1.         0.        ]
[0.11111111 0.13684211 0.         1.         0.        ]
[0.         0.         0.         1.         0.        ]
[0.55555556 0.73684211 0.         1.         0.        ]
[0.22222222 0.26315789 0.         1.         0.        ]
[0.33333333 0.42105263 0.         1.         0.        ]
[0.66666667 0.84210526 0.         1.         0.        ]
[0.11111111 0.15789474 0.         1.         0.        ]
[0.44444444 0.63157895 0.         1.         0.        ]
[0.         0.05263158 0.         1.         0.        ]
[0.22222222 0.31578947 0.         1.         0.        ]
[0.77777778 0.89473684 0.         1.         0.        ]
[0.33333333 0.47368421 0.         1.         0.        ]
[0.11111111 0.17894737 0.         1.         0.        ]
[0.55555556 0.68421053 0.         1.         0.        ]
[0.         0.07368421 0.         1.         0.        ]
[0.44444444 0.52631579 0.         1.         0.        ]
[0.22222222 0.24210526 0.         1.         0.        ]
[0.88888889 0.94736842 0.         1.         0.        ]
[0.33333333 0.45263158 0.         1.         0.        ]
[0.11111111 0.12631579 0.         1.         0.        ]
[0.66666667 0.78947368 0.         1.         0.        ]
[0.         0.04210526 0.         1.         0.        ]
[0.44444444 0.6        0.         1.         0.        ]
[0.22222222 0.28421053 0.         1.         0.        ]
[1.         1.         0.         1.         0.        ]
[0.11111111 0.15789474 0.         0.         1.        ]
[0.         0.05263158 0.         0.         1.        ]
[0.22222222 0.31578947 0.         0.         1.        ]
[0.33333333 0.47368421 0.         0.         1.        ]
```

[0.44444444 0.63157895 0.   0.   1.   ]

[0.11111111 0.13684211 0.   0.   1.   ]

[0.   0.03157895 0.   0.   1.   ]

[0.55555556 0.73684211 0.   0.   1.   ]

[0.22222222 0.26315789 0.   0.   1.   ]

[0.33333333 0.42105263 0.   0.   1.   ]

[0.66666667 0.84210526 0.   0.   1.   ]

[0.11111111 0.17894737 0.   0.   1.   ]

[0.44444444 0.57894737 0.   0.   1.   ]

[0.   0.07368421 0.   0.   1.   ]

[0.22222222 0.36842105 0.   0.   1.   ]

[0.77777778 0.89473684 0.   0.   1.   ]

[0.33333333 0.45263158 0.   0.   1.   ]

[0.11111111 0.21052632 0.   0.   1.   ]

[0.55555556 0.68421053 0.   0.   1.   ]

[0.   0.10526316 0.   0.   1.   ]

[0.44444444 0.52631579 0.   0.   1.   ]

[0.22222222 0.28421053 0.   0.   1.   ]

[0.88888889 0.94736842 0.   0.   1.   ]

[0.33333333 0.49473684 0.   0.   1.   ]

[0.11111111 0.14736842 0.   0.   1.   ]

[0.66666667 0.81052632 0.   0.   1.   ]

[0.   0.06315789 0.   0.   1.   ]

[0.44444444 0.55789474 0.   0.   1.   ]

[0.22222222 0.34736842 0.   0.   1.   ]

[1.   1.   0.   0.   1.   ]

[0.11111111 0.10526316 1.   0.   0.   ]

[0.   0.05263158 1.   0.   0.   ]

[0.22222222 0.26315789 1.   0.   0.   ]

[0.33333333 0.42105263 1.   0.   0.   ]

[0.44444444 0.57894737 1.   0.   0.   ]

```
 [0.11111111 0.13684211 1.         0.         0.        ]
 [0.         0.03157895 1.         0.         0.        ]
 [0.55555556 0.73684211 1.         0.         0.        ]
 [0.22222222 0.31578947 1.         0.         0.        ]
 [0.33333333 0.47368421 1.         0.         0.        ]
 [0.66666667 0.78947368 1.         0.         0.        ]
 [0.11111111 0.15789474 1.         0.         0.        ]
 [0.44444444 0.63157895 1.         0.         0.        ]
 [0.         0.07368421 1.         0.         0.        ]
 [0.22222222 0.36842105 1.         0.         0.        ]
 [0.77777778 0.84210526 1.         0.         0.        ]
 [0.33333333 0.45263158 1.         0.         0.        ]
 [0.11111111 0.17894737 1.         0.         0.        ]
 [0.55555556 0.68421053 1.         0.         0.        ]
 [0.         0.10526316 1.         0.         0.        ]
 [0.44444444 0.52631579 1.         0.         0.        ]
 [0.22222222 0.28421053 1.         0.         0.        ]
 [0.88888889 0.89473684 1.         0.         0.        ]
 [0.33333333 0.49473684 1.         0.         0.        ]
 [0.11111111 0.21052632 1.         0.         0.        ]
 [0.66666667 0.81052632 1.         0.         0.        ]
 [0.         0.06315789 1.         0.         0.        ]
 [0.44444444 0.55789474 1.         0.         0.        ]
 [0.22222222 0.34736842 1.         0.         0.        ]
 [1.         0.94736842 1.         0.         0.        ]]
0.9504220042269662
model saved
mms saved
```

P5.py

C:\ml\L19>python p5.py

model ready

mms ready

enter age 3

enter kms 400000

1 for Mahindra , 2 for Maruti and 3 for Tata 3

[176444.44444444]

p2_usedcar_pricepredictor_app